An Empirical Study of Training and Post-Training Strategies for VGG16–GeM Image Retrieval

Author 1 Author 2 Author 3

***Abstract*—Improving an image-retrieval pipeline requires separating encoder quality from the effects and costs of search-time processing. We study a VGG16–GeM reproduction and two fine-tuned encoders with augmentation, with and without a lightweight spatial gate. Training uses a structure-from-motion subset containing 15,953 training images and 739 validation images. The historical reproduction remains the strongest available clean checkpoint, whereas both new runs show late-training validation deterioration. We then freeze both new encoders at epoch 100 and evaluate query expansion, local and geometric re-ranking, two-encoder fusion, and graph diffusion on the legacy Oxford5k and Paris6k protocols. Query robustness is assessed using six corruption families at three severity levels, with clean galleries. For the augmentation-only encoder, alpha-weighted query expansion increases mean corrupted retrieval performance across the two datasets from 66.82 to 69.34 percent, while geometry-verified expansion reaches 71.05 percent. The latter requires median query times of 8.48 and 5.82 seconds on Oxford and Paris, respectively. Fusion does not improve upon the stronger encoder. An implementation audit reproduces the severe archived diffusion degradation with dense query seeds, while fixed top-20 seeds recover Oxford clean mAP from 23.67 to 76.75 percent on the same union graph. The study identifies feedback quality, checkpoint choice and diffusion initialization as practical determinants of retrieval performance; its scope and experimental limitations are stated separately.**

***Index Terms*—instance image retrieval, generalized-mean pooling, query expansion, geometric verification, robustness evaluation**

I. Introduction

Instance image retrieval ranks gallery images depicting the same particular object or landmark as a query. A global descriptor enables inexpensive similarity search, but its usefulness depends on both the learned representation and the processing applied after the initial ranking. Fine-tuning can adapt a classification backbone to retrieval, while query expansion, local verification, fusion, and graph propagation offer additional ways to alter rankings without updating encoder weights.

The VGG16–GeM pipeline provides an established setting for examining these choices. VGG networks supply convolutional representations [1], and generalized-mean pooling (GeM) supports compact descriptors learned from structure-from-motion (SfM) supervision [2]. However, adding augmentation, attention, or more epochs does not by itself establish an improvement. Likewise, a search method that helps a fixed final checkpoint need not outperform a stronger historical checkpoint. Mixing these comparisons can obscure what an experiment actually demonstrates.

We examine three available training histories: a baseline reproduction, denoted B0, an augmentation-only run (R1), and an augmentation-plus-spatial-gate run (R2). The training comparison is historical rather than a fully controlled ablation. We separately conduct a fixed-encoder search study using the two new epoch-100 checkpoints, re-extracted with the same FP32 inference protocol. This separation preserves the meaning of both comparisons.

Our study addresses three questions. *RQ1:* How do validation-selected and final checkpoints compare, and what evidence supports the training modifications? *RQ2:* Which post-training methods improve clean and corrupted-query retrieval for the same fixed encoder? *RQ3:* How much query-time computation is required for these gains?

The contributions are empirical: (1) an audit of training configuration, data splits, and checkpoint selection; (2) an evaluation of four search families and geometry-verified query expansion on two encoders under a fixed corruption protocol; and (3) a comparison of retrieval-only and direct end-to-end latency, retaining unsuccessful methods and exploratory sensitivity results. The work evaluates combinations of established components; it does not introduce a new pooling, attention, or geometric-verification algorithm.

II. Related Work

A. Learned Global Representations

Radenović *et al.* use SfM reconstructions to obtain matching and non-matching training examples without manual retrieval labels, and introduce trainable GeM pooling [2]. We adopt this descriptor-learning framework on a prepared subset, retaining a single global descriptor and disabling whitening and multi-scale inference. Our results therefore should not be interpreted as a reproduction of every component or headline score of the original system.

Spatial attention is another way to alter feature aggregation. CBAM combines channel and spatial attention [3]. Our second run uses only a small spatial gate inspired by channel-mean and channel-max aggregation, with identity-preserving initialization and a scaled sigmoid. It is not a full CBAM implementation.

Modern visual backbones broaden the representation choices. Vision Transformer (ViT) processes images as sequences of patches [4], while DINOv2 learns transferable visual features through large-scale self-supervision [5]. These approaches provide relevant alternatives to the convolutional encoder studied here. Our VGG16 experiments examine

checkpoint selection and search processing within one established descriptor family; they do not rank modern backbones.

B. Feedback, Geometry, and Ranking

Visual query expansion uses initially retrieved images to refine the query. Chum *et al.* combine feedback with spatial verification to limit false-positive expansion [6]. We evaluate descriptor-level averaging, similarity weighting, a cosine gate, and a geometry-filtered feedback pipeline. RootSIFT applies a normalization and square-root transformation to local SIFT descriptors [7]; it supplies the local evidence in our re-ranking and verification stages. Homographies are estimated using RANSAC [8].

Reciprocal rank fusion (RRF) combines rank positions rather than requiring matching score scales [9]. Diffusion can exploit neighborhood structure; Iscen *et al.* study propagation over regional CNN representations [10]. Our graph is built from one global descriptor per gallery image, so a failure here is not evidence against their regional method.

DELG unifies learned local and global features in one network, combining GeM-based global retrieval with attentive local representations [11]. SuperGlobal instead refines global representations for both retrieval and re-ranking [12]. Together they motivate evaluating the cost of local matching against global-only feedback. Our RootSIFT verification and residual QE are simpler reference pipelines, with their storage and query costs measured explicitly.

C. Benchmark Protocols

Oxford and Paris originate from landmark-retrieval research [13], [14]. Revised annotations and evaluation protocols were subsequently introduced [15]. We use the available *legacy* ground truth with 55 queries per dataset. Results are consequently not interchangeable with revisited Easy, Medium, or Hard scores. Our corruption suite is a deterministic extension of these query images, not a newly annotated independent benchmark.

III. Methods

A. Encoder and Tuple Objective

For nonnegative convolutional activations $x_{c,u}$ at channel c and position $u \in \mathcal{U}$, GeM computes

$$g_c = \left(\frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} x_{c,u}^p \right)^{1/p}, \qquad \mathbf{z} = \frac{\mathbf{g}}{\|\mathbf{g}\|_2}. \tag{1}$$

A shared learnable p starts at 3. VGG16 features, without the classifier and final pooling, produce a 512-dimensional normalized descriptor. The initial ranking uses cosine score $s_i = \mathbf{z}_q^\top \mathbf{z}_i$.

Each training tuple contains a query q, one metadata-defined positive p, and five negatives n_j. The implemented objective is

$$\mathcal{L} = \tfrac{1}{2} d(q,p)^2 + \tfrac{1}{2} \sum_{j=1}^{5} [m - d(q,n_j)]_+^2, \quad m = 0.7, \tag{2}$$

where $[a]_+ = \max(a, 0)$ and d is Euclidean descriptor distance. The implementation adds 10^{-6} to coordinate differences before computing the norm for numerical stabilization. Positive relations come from explicit metadata pairs; cluster membership alone does not assign a positive label. Hard-negative mining uses current clean descriptors, excludes the query cluster, and selects negatives from distinct clusters.

B. Training Modifications

R1 adds independent photometric and degradation transforms to the images used in the training forward pass. With probability 0.8, brightness and contrast factors are drawn from $[0.8, 1.2]$ and color from $[0.85, 1.15]$, in random order. Gaussian blur is applied with probability 0.15 and radius $[0.2, 1.0]$ pixels; JPEG encoding with probability 0.2 and integer quality 55–95; and erasing with probability 0.15, area 1–5%, and aspect ratio 0.5–2. Erasing fills zero after normalization, corresponding to the normalization mean in RGB. Clean mining and validation do not use these transforms.

R2 adds a spatial gate before GeM. For feature tensor X, the gate is

$$A(X) = 2\sigma(\mathrm{Conv}_{7\times 7}[\mathrm{mean}_c\, X; \max_c X]). \tag{3}$$

The two-input, one-output convolution has 99 parameters including bias. Zero-initialized weights and bias give $A(X) = 1$, so the initial gated tensor equals X. The output descriptor dimensionality remains unchanged.

C. Query Expansion and Feedback Selection

Let $\mathcal{F}$ be positive-cosine candidates from the first top-k ranking. We compute

$$\mathbf{q}' = \mathrm{norm}_2\left(\mathbf{q} + \lambda \frac{\sum_{i \in \mathcal{F}} w_i \mathbf{z}_i}{\sum_{i \in \mathcal{F}} w_i} \right), \quad w_i = \max(s_i, 0)^\alpha. \tag{4}$$

AQE uses $\alpha = 0$; weighted QE uses $\alpha = 3$. We fix $k = 10$ and $\lambda = 0.5$ for one feedback iteration. Gated QE additionally requires $s_i \geq 0.5$ and at least three accepted candidates, otherwise returning the original query. Setting $\lambda = 0$ also returns the baseline. The residual form in (4) is stated explicitly because conventions for mixing query and feedback differ.

D. Local and Geometric Evidence

We extract up to 2,000 RootSIFT keypoints per image. Matching uses a 0.75 nearest-neighbor ratio test and mutual nearest-neighbor consistency. Homography RANSAC uses a 3-pixel threshold, confidence 0.995, and at most 2,000 iterations. A valid result requires at least eight inliers and a finite, full-rank homography with condition number below 10^{12}.

For a top-50 candidate with M_i matches and I_i inliers, local re-ranking uses evidence $e_i = \min(M_i/50, 1)$. Geometric re-ranking instead uses

$$e_i = \frac{I_i}{\max(M_i, 1)} \min(I_i/20, 1) \tag{5}$$

when geometry is valid, and zero otherwise. The combined score is $(1 - \eta)(s_i + 1)/2 + \eta e_i$, with $\eta = 0.5$. Only the candidate prefix changes; the remaining ranking is retained.

Invalid query features or no valid geometric candidate trigger a logged fallback where applicable.

Verified QE checks the first 50 global candidates, selects at most ten with valid geometry and positive cosine, and applies (4). Fewer than three valid feedback images returns the original query. Search is then repeated over the entire gallery, without a second re-ranking stage. Relevance and junk labels never select feedback.

E. Fusion and Diffusion

Score fusion averages the two encoders' cosine scores after image-ID alignment. RRF uses both full rankings:

$$S_{\mathrm{RRF}}(i) = \sum_{r=1}^{2} \frac{1}{60 + \mathrm{rank}_r(i)}. \tag{6}$$

We do not average raw embeddings across encoders.

Each encoder also defines a gallery graph with $k = 20$, no self-edges, and directed weights $\max(s_{ij}, 0)^3$. Union and mutual graphs use the elementwise maximum and minimum with the transposed affinity, respectively. Let $S = D^{-1/2} W D^{-1/2}$, using zero inverse degree for isolated nodes. Initial query scores y_i are positive-cosine cubes normalized to unit sum. We iterate

$$\mathbf{f}^{(t+1)} = (1 - \beta)\mathbf{y} + \beta S \mathbf{f}^{(t)}, \quad \beta = 0.9, \tag{7}$$

starting at $\mathbf{f}^{(0)} = \mathbf{y}$, stopping at relative L2 change below 10^{-6} or 100 iterations. Non-finite results or absent initial mass fall back to cosine. Labels are not used in graph construction or propagation.

The archived grid above uses a dense positive-similarity seed. A subsequent implementation audit evaluates a fixed sparse alternative: retain only the first 20 query neighbors in $\mathbf{y}$, with graph construction, exponent and β unchanged. For this diagnostic, both dense and sparse variants use tolerance 10^{-12} and at most 1,000 iterations, and an independent linear solver checks the result. The sparse seed follows the support convention in the reference implementation [10]; it is evaluated separately from the archived main grid.

IV. Experimental Setup

A. Training Data and Validation

The prepared SfM subset contains 15,953 training images from 100 clusters and 34,337 explicit query–positive pairs. Validation contains 739 images from 20 clusters and 181 pairs. Image identifiers and clusters are disjoint between these partitions. We have not established semantic landmark disjointness between SfM and the retrieval benchmarks.

Each epoch samples 2,000 training tuples. The requested mining pool of 20,000 is clipped to the 15,953 available training images. Thus, an epoch does not enumerate all positive pairs. Five tuple losses are summed before each Adam update, giving 400 updates per epoch. Although each update contains 35 image positions, duplicate images are possible. The microbatch limit of eight applies only to same-shape forward groups; it does not change the optimizer's five-tuple batch. Tuple fusion is one in the new runs.

TABLE I
Training configuration. LR denotes base learning rate before scheduling. B0 is a historical reproduction.

Setting	B0	R1 / R2
Optimizer	Adam	Adam
Backbone LR	10^{-6}	10^{-6}
GeM LR	10^{-5}	10^{-5}
Gate LR	—	— / 10^{-4}
Backbone weight decay	10^{-6}	10^{-6}
GeM / gate weight decay	0 / —	0 / 0
Training seed	42	42
Image long-edge cap	1024	1024
Tuples / update	5	5
Negatives / tuple	5	5
Completed epochs	14	100 / 100
Backbone train precision	FP32	BF16 autocast

Validation uses no augmentation or gradient updates. With batch size five and the final incomplete batch dropped, 180 of the 181 validation tuples are processed. For the two new runs, validation pair sampling is seeded consistently, but mined negatives still depend on the current encoder. The selection criterion is validation contrastive loss, not benchmark mAP. Pair metadata alone does not provide exhaustive relevance labels for a retrieval development set.

B. Optimization and Checkpoint Identity

Table I summarizes the settings. Adam [16] uses a pre-epoch schedule $\mathrm{LR}(e) = \mathrm{LR}_0 \exp(-0.01e)$, beginning at $e = 1$. The new runs retain FP32 for GeM, descriptor normalization, loss, weights, and Adam state. They use four workers, extraction window eight, prefetch factor two, and no image cache. Training uses Colab; the new runs were configured for an A100 40-GB GPU. Runtime differences prevent an algorithm-only speed comparison with B0.

We distinguish two evidence tracks. The *historical checkpoint comparison* uses saved clean evaluations: B0 best at epoch 12 and last available at 14, R1 best at 27, and R2 best at 28. The archived new-run best evaluations use BF16 backbone inference. The *post-training comparison* uses both latest epoch-100 checkpoints, re-extracted in FP32 with TF32 disabled. B0 historical evaluations are FP32. Consequently, cross-track differences are not a controlled precision comparison.

The R1 epoch-27 weight file is absent from the supplied archive, although its saved evaluation and history remain available. Its clean scores are reported as historical evidence, not as a newly repeated inference. Both final checkpoints and the B0 best checkpoint are available. Archive, source-checkpoint, exported-weight, and metadata hashes identify the inputs; exported final weights were checked against the source model before extraction.

C. Retrieval Protocol and Corruptions

Oxford5k contains 5,063 gallery images and Paris6k 6,392, each with 55 query regions. We retain the legacy bounding boxes, relevant-image labels, junk handling, and trapezoidal

TABLE II
QUERY CORRUPTIONS AT SEVERITIES 1/2/3. L IS THE RESIZED QUERY LONG EDGE; THE GALLERY STAYS CLEAN.

Family	Severity parameters
Darkening	RGB factors 0.8 / 0.6 / 0.4
Contrast	Factors about channel mean: 0.8 / 0.6 / 0.4
Blur	Gaussian σ/L: 0.001 / 0.002 / 0.004
Noise	Gaussian RGB std.: 0.02 / 0.05 / 0.08
JPEG	Quality 80 / 50 / 25; subsampling 2
Occlusion	Area fractions 0.05 / 0.10 / 0.20; query RGB-mean fill

TABLE III
HISTORICAL CLEAN MAP (%). BEST MEANS MINIMUM VALIDATION LOSS. R1/R2 BEST USE ARCHIVED BF16-BACKBONE EVALUATIONS; FINAL ROWS ARE THE NEW FP32 COSINE BASELINES.

Checkpoint	Epoch	Oxford	Paris
B0 best	12	81.06	77.12
B0 last available	14	80.77	76.08
R1 best	27	79.53	71.93
R2 best	28	80.49	72.76
R1 final	100	74.00	64.29
R2 final	100	71.37	62.72

AP evaluator. Image-source queries may also occur in the gallery. AP is averaged over queries to form mAP. We additionally retain P@10 and relevant Recall@20/50/100 after junk removal. These are legacy scores, not revisited-protocol results [15].

Inference uses one scale and no whitening. RGB images preserve aspect ratio under a long-edge cap of 1024. Query crops retain the upstream relative-resize rule: their cap is scaled by the ratio of crop to original long edge, rather than enlarging every crop to 1024. Values in $[0, 1]$ are normalized by mean $(0.485, 0.456, 0.406)$ and standard deviation $(0.229, 0.224, 0.225)$.

Corruptions in Table II are applied after crop/resize and before normalization. Blur uses a minimum sigma of 0.1 pixels. Noise and occlusion each have three fixed realizations per severity; other families have one. This produces 30 corrupted views plus clean per query: 3,410 query instances or 6,820 forwards across the two encoders. Corruption seed 20261030 is independent of training seed 42. Pixel checksums enforce consistency between global and local features. Noise is absent from the training augmentation; the remaining families overlap in part but differ in generation or strength.

For $C = 18$ family–severity conditions and their realization counts R_c, we report

$$\mathrm{mAP}_{\mathrm{corr}} = \frac{1}{C} \sum_{c=1}^{C} \frac{1}{R_c} \sum_{r=1}^{R_c} \mathrm{mAP}_{c,r}. \tag{8}$$

Dataset-macro results average Oxford and Paris equally. Worst-condition mAP is the minimum of the 18 realization-averaged condition scores. Drop is clean minus corrupted mAP in percentage points. Repeated views of the same query are not independent training replications.

D. Frozen Parameters, Sensitivity, and Cost

Main parameters are fixed across datasets and corruption conditions. Clean-only, one-axis sensitivity varies QE $k \in \{5, 10, 20\}$, $\alpha \in \{1, 3, 5\}$, $\lambda \in \{0.25, 0.5, 1\}$, gate $\tau \in \{0.4, 0.5, 0.6\}$; re-ranking $K \in \{20, 50, 100\}$ and $\eta \in \{0.25, 0.5, 0.75\}$; RRF $c \in \{10, 60, 100\}$; and diffusion $k \in \{10, 20, 40\}$ and $\beta \in \{0.5, 0.9, 0.99\}$. Main settings supply the reference, with 100 additional model–method–configuration–dataset rows. No separate labeled retrieval development set is available. We call all search results exploratory and do not promote a sensitivity maximum to the main configuration.

FP32 descriptor extraction used a Kaggle Tesla T4, PyTorch 2.11.0+cu128, NumPy 2.1.3, and Pillow 13.0.0. Local processing used OpenCV 4.14.0; CPU jobs used SciPy 1.16.3. CPU allocation can differ between Kaggle jobs, so timings describe the measured deployment rather than a hardware-normalized algorithm benchmark.

Retrieval-only latency assumes cached descriptors. Direct end-to-end latency includes decoding, query crop/resize, corruption when applicable, encoding, search, and query local extraction/matching for local methods. Gallery descriptors, gallery local features, and graphs are offline assets. Timing uses batch one, warm-up, and GPU synchronization. For each model/dataset/method, 28 samples cover clean and severity-three queries from all six families, with two queries per condition and two repeats. Across 40 groups, this gives 1,120 direct samples. We report medians and retain p95 in the released tables; independently measured medians are not additive component costs.

V. RESULTS AND DISCUSSION

A. RQ1: Checkpoint Selection and Training Changes

B0 best remains strongest on both clean benchmarks (Table III), reaching 81.06% on Oxford and 77.12% on Paris. The best available historical evaluations for R1 reach 79.53% and 71.93%; R2 reaches 80.49% and 72.76%. The gate run is higher at these selected checkpoints, but this ordering reverses at epoch 100. The gate advantage therefore depends on checkpoint selection, and both new runs remain below the selected B0 checkpoint.

The histories show late-training deterioration (Fig. 1). R1 has minimum validation loss 0.100764 at epoch 27, rising to 0.123108 at epoch 100; R2 rises from 0.100519 at epoch 28 to 0.119673. Meanwhile, training loss falls to approximately 0.0288 and 0.0280. Combined with lower final retrieval scores, this is consistent with overfitting. The training and validation trajectories separate as learning continues.

The archived B0 best epoch is 12, selected by validation, not by taking the best score separately on each benchmark. For example, B0's Paris score at epoch 8 is higher than at epoch 12, but substituting it would mix selection criteria. We preserve the declared checkpoint rule. Similarly, a lower validation loss across two runs is not automatically a calibrated estimate of better retrieval, particularly with changing mined negatives.

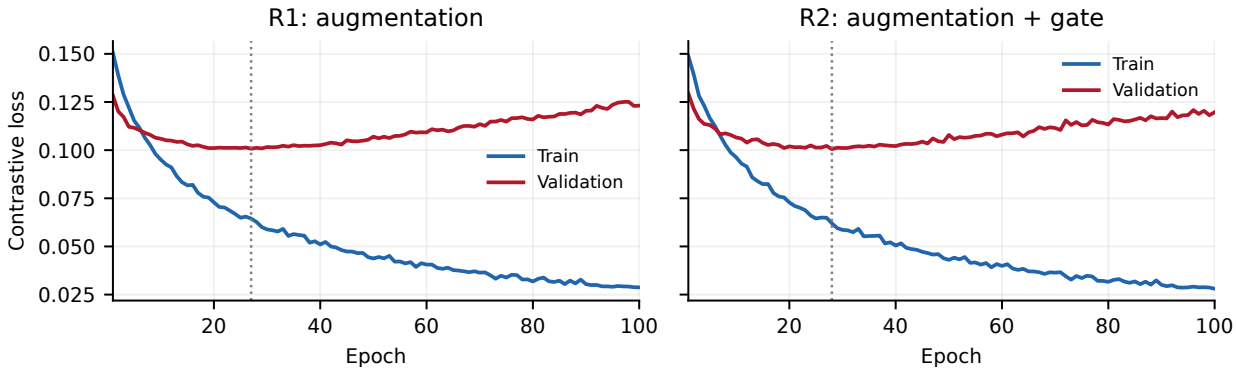


Fig. 1. Training and validation contrastive loss for the two new runs. Dotted lines mark validation-selected epochs 27 and 28. Validation negatives depend on the current encoder; curves show the implemented monitoring objective.

B. RQ2: Search Improvements for Fixed Encoders

a) Expansion provides low-cost gains: AQE and weighted QE improve both clean and corrupted mAP for both final encoders (Table IV). For R1, weighted QE increases Oxford clean from 74.00% to 76.45% and Paris clean from 64.29% to 65.89%. Equal-dataset corrupted mAP rises from 66.82% to 69.34%, a 2.52-percentage-point gain. AQE and weighted QE give similar improvements.

The cosine gate opens for every measured query view and accepts ten feedback images, causing gated QE to equal weighted QE. Identical scores identify an inactive selection constraint.

b) Geometry is most useful as a feedback filter: For R1, local-match re-ranking reaches 78.53% Oxford clean, exceeding the geometric-score re-ranker at 76.68%. Geometric verification is therefore not uniformly stronger when used only to reorder a short prefix. In contrast, verified QE reaches 78.94% and 66.97% clean on Oxford and Paris, and 76.28% and 65.82% under corruption. Its corrupted dataset-macro mAP is 71.05%, or 4.23 points above cosine. For R2, verified QE also gives the highest main-method mAP in all four columns: 76.85%, 74.25%, 65.10%, and 64.31%.

Verified feedback is not always available. Across clean and corrupted views, its gate opens for 88.39% of R1 Oxford cases and 92.96% of Paris cases. The mean feedback counts, including closed gates, are 6.87 and 8.56. Fallbacks are retained rather than removing difficult queries from evaluation. The logged feedback counts expose how often geometric filtering can operate.

c) Fusion does not beat the stronger encoder: Neither score fusion nor RRF exceeds R1 cosine on either clean dataset or their corrupted means (Table V). Score fusion obtains 73.55% Oxford clean and 64.04% Paris clean, compared with 74.00% and 64.29% for R1. The additional encoder and combination step provide no accuracy gain for this pair.

d) Dense query seeds explain the archived diffusion collapse: Mutual diffusion reduces R1 Oxford clean from 74.00% to 48.95%, and union diffusion to 23.67%. Paris also deteriorates. This is not primarily explained by numerical failure: aggregate diagnostics report full convergence for both R1 graphs, zero fallback, and mean iteration counts approximately 63–93. R2 Oxford mutual diffusion reaches convergence in 99.77% of views, with the remaining views retained at the iteration cap.

An implementation audit changes this interpretation. Holding the cached graph, descriptors and β fixed, top-20 query seeds raise R1 Oxford clean mAP to 76.75% on the union graph and 77.46% on the mutual graph, compared with 74.00% cosine. Paris mutual diffusion reaches 77.26%, compared with 64.29% cosine. Dense-seed replay reproduces the original collapse. Fixed-point solutions agree with independent conjugate-gradient solutions within 1.5×10^{-11} relative error on three fixed queries per case; hand-checkable graphs also agree with direct linear solves. The evidence identifies seed support as a consequential implementation choice, rather than supporting a claim that diffusion generally degrades retrieval. The archived table retains its original dense-seed measurements for traceability.

Replaying this single fixed mutual/top-20 configuration on all 31 archived views produces 6,820 new per-query AP records (Table VI). For R1, corrupted dataset-macro mAP is 76.27%, compared with 66.82% cosine and 71.05% verified QE in the archived study. This extends the seed-support diagnosis to the prescribed corruptions without additional encoder inference. It adds no new end-to-end latency measurement.

C. Corruption Severity and Exploratory Sensitivity

Figure 2 shows that severe blur is particularly damaging to R1. Its worst condition is blur severity three on both datasets: cosine mAP is 48.33% Oxford and 54.71% Paris. Verified QE raises those same condition scores to 60.97% and 58.60%, respectively. Improvements remain conditional on the starting encoder and the particular query transformations. A smaller clean-to-corrupted drop alone would not demonstrate a better system if absolute mAP were lower.

The suite partially overlaps training augmentation but is not identical. In particular, Gaussian noise is not included in the saved augmentation policy. Occlusion includes cases

TABLE IV

ARCHIVED POST-TRAINING RESULTS, MAP (%). C DENOTES CLEAN AND R THE MEAN OVER 18 CORRUPTED-QUERY CONDITIONS. BOTH ENCODERS ARE FIXED AT EPOCH 100 AND USE FP32 EXTRACTION. EACH METHOD USES ITS ENCODER'S OWN COSINE BASELINE; DIFFUSION HERE USES THE ORIGINAL DENSE SEEDS.

	R1: augmentation				R2: augmentation + gate			
	Oxford5k		Paris6k		Oxford5k		Paris6k	
Method	C	R	C	R	C	R	C	R
Cosine	74.00	70.80	64.29	62.84	71.37	68.15	62.72	61.59
AQE	76.41	73.81	65.90	64.84	73.64	70.86	64.06	63.36
α-QE	76.45	73.86	65.89	64.83	73.63	70.90	64.06	63.36
Gated QE	76.45	73.86	65.89	64.83	73.63	70.90	64.06	63.36
Local re-rank	78.53	75.08	65.42	64.01	76.55	73.35	63.92	62.83
Geometric re-rank	76.68	73.46	64.83	63.40	74.98	71.84	63.33	62.19
Verified QE	78.94	76.28	66.97	65.82	76.85	74.25	65.10	64.31
Mutual diffusion	48.95	47.32	53.11	52.26	51.15	49.63	52.55	52.01
Union diffusion	23.67	22.10	45.61	45.15	25.79	24.13	46.15	46.05

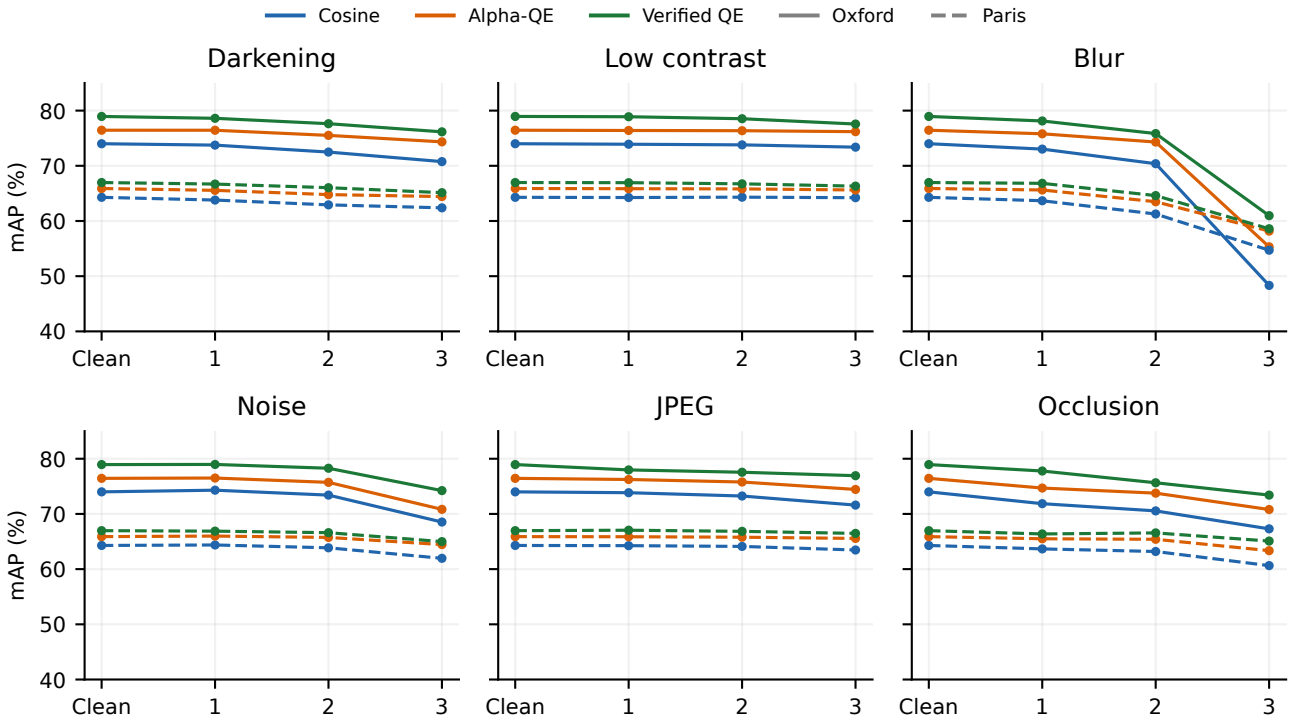


Fig. 2. R1 retrieval under increasing query corruption. Solid curves: Oxford; dashed: Paris. Blue: cosine; orange: weighted QE; green: verified QE. Each non-clean point averages the prescribed realizations. The gallery is always clean.

TABLE V

TWO-ENCODER FUSION, MAP (%). BOTH USE R1+R2 EPOCH 100; COLUMN ORDER MATCHES TABLE IV.

	Oxford		Paris	
Method	C	R	C	R
Score fusion	73.55	70.40	64.04	62.77
RRF	73.43	70.25	64.11	62.80

TABLE VI

REVISED MUTUAL DIFFUSION WITH FIXED TOP-20 SEEDS, MAP (%). C: CLEAN; R: MEAN OVER 18 CORRUPTED CONDITIONS. THE SAME CACHED EPOCH-100 DESCRIPTORS AND GRAPHS ARE REUSED.

	Oxford		Paris	
Encoder	C	R	C	R
R1	77.46	76.21	77.26	76.34
R2	73.80	72.45	77.79	77.06

where identifying content is removed; this stress test covers the prescribed transformations.

Clean sensitivity contains 100 additional grouped evaluations, representing 5,500 per-query AP values. For instance,

R1 weighted QE with $\lambda = 1$ reaches 77.30% Oxford and 66.56% Paris clean, above the frozen $\lambda = 0.5$ setting. Local re-ranking with $K = 100$ and $\eta = 0.5$ reaches 78.85% and 66.94%. The supplement retains all settings and outcomes; the main corruption experiment keeps its frozen configuration.

D. RQ3: Quality–Latency Trade-offs

QE adds approximately 3–5 ms to retrieval-only processing for R1. Direct weighted-QE medians are 75.55 ms on Oxford and 87.34 ms on Paris. Verified QE has medians 8.48 s and 5.82 s; local matching and geometric processing dominate this scale of cost. These methods occupy substantially different deployment regimes (Fig. 3).

The direct cosine medians for R1 are 85.52 and 109.53 ms, larger than the corresponding weighted-QE medians despite QE's additional operation. Independent timing samples show execution variability; retrieval-only timings expose the added expansion work more directly.

Offline computation also matters. The shared local-feature cache reports approximately 2.56 GB of stored data. Its recorded builds take 2,464 s for Oxford and 3,049 s for Paris, covering gallery and query-view features. Query local features are nevertheless recomputed in the direct timer so cached queries do not hide online costs. Both fusion encoders must also be executed: direct score-fusion medians are 147.17 ms Oxford and 160.30 ms Paris. Its extra encoding cost does not yield an accuracy improvement here.

E. Reproducibility

The main grid has 20 ranking configurations per dataset: nine per encoder and two fusion rules. Across 110 queries and 31 views this yields 68,200 main AP records, with no missing cells in the saved coverage audit. Together with sensitivity, the saved per-query records total 73,700. Descriptors, rankings, image IDs, pixel checksums, checkpoint hashes, graph diagnostics, and job completion states identify the evaluated artifacts. Smoke checks cover export/reload consistency, finite descriptors, perfect/junk evaluator cases, QE identity and closed-gate behavior, single-ranker fusion, prefix-preserving re-ranking, and graph numerical behavior.

VI. Limitations

Only one training seed is available. Corrupted views of the same query and related landmark queries are not independent training replications. Training histories differ in budget, precision and runtime, and R1's validation-selected weights are absent from the supplied archive. These constraints limit attribution of differences to augmentation or attention. Evaluation uses legacy Oxford/Paris annotations; comparison with modern systems requires ROxford/RParis Medium/Hard under a shared protocol. The sparse-seed audit is a post-review implementation diagnosis on the existing benchmarks, rather than an independent validation set. Search sensitivity is descriptive: no test-set maximum replaces the frozen main settings. A separately labeled retrieval development set is still needed for parameter selection. The original B0 record lacks a matched corruption study, and direct latency samples are small. Revised sparse diffusion has no new end-to-end timing measurement.

VII. Conclusion

This empirical study separates training history from fixed-encoder search improvements in a VGG16–GeM retrieval pipeline. The available baseline reproduction remains strongest among the historical clean checkpoints, while the two new runs exhibit late-training validation deterioration. For their epoch-100 encoders, query expansion improves retrieval at low additional search cost. Geometry-verified expansion produces the best archived main post-training scores but requires seconds per query. Fusion fails to improve the stronger encoder. An implementation audit traces the severe archived diffusion degradation to dense query seeding: fixed top-20 seeds recover clean mAP above cosine on both benchmarks.

The practical findings are the accuracy–cost trade-off of feedback filtering and the importance of faithful diffusion initialization. Validation-selected checkpoints, consistent inference and separate accounting for offline preparation and online latency make these comparisons useful. The released source, checks and raw numerical outputs support tracing each result to its implementation.

References

[1] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *International Conference on Learning Representations*, 2015. [Online]. Available: https://arxiv.org/abs/1409.1556

[2] F. Radenović, G. Tolias, and O. Chum, "Fine-tuning CNN image retrieval with no human annotation," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 41, no. 7, pp. 1655–1668, 2019.

[3] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *European Conference on Computer Vision*, 2018. [Online]. Available: https://arxiv.org/abs/1807.06521

[4] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16x16 words: Transformers for image recognition at scale," in *International Conference on Learning Representations*, 2021. [Online]. Available: https://arxiv.org/abs/2010.11929

[5] M. Oquab, T. Darcet, T. Moutakanni *et al.*, "DINOv2: Learning robust visual features without supervision," *arXiv preprint arXiv:2304.07193*, 2023. [Online]. Available: https://arxiv.org/abs/2304.07193

[6] O. Chum, J. Philbin, J. Sivic, M. Isard, and A. Zisserman, "Total recall: Automatic query expansion with a generative feature model for object retrieval," in *IEEE International Conference on Computer Vision*, 2007, pp. 1–8. [Online]. Available: https://www.robots.ox.ac.uk/~vgg/publications/2007/Chum07b/chum07b.pdf

[7] R. Arandjelović and A. Zisserman, "Three things everyone should know to improve object retrieval," in *IEEE Conference on Computer Vision and Pattern Recognition*, 2012, pp. 2911–2918.

[8] M. A. Fischler and R. C. Bolles, "Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography," *Communications of the ACM*, vol. 24, no. 6, pp. 381–395, 1981. [Online]. Available: https://www.sri.com/wp-content/uploads/2021/12/ransac-publication.pdf

[9] G. V. Cormack, C. L. A. Clarke, and S. Büttcher, "Reciprocal rank fusion outperforms condorcet and individual rank learning methods," in *International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2009, pp. 758–759.

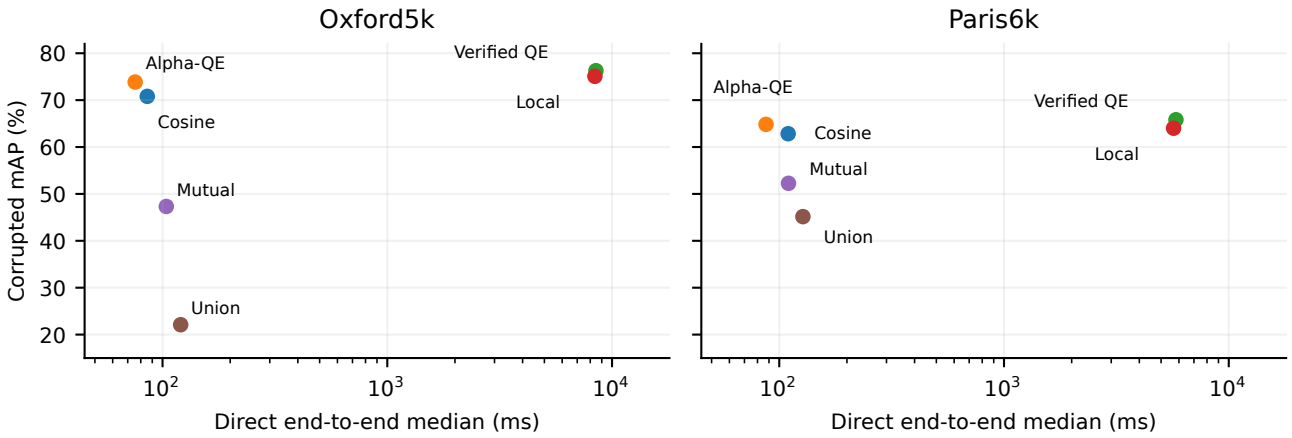


Fig. 3. R1 corrupted mAP versus measured direct end-to-end median query latency (logarithmic horizontal axis). Quality uses the complete corruption suite; latency uses the predefined 28-sample subset. Offline gallery preparation is excluded.

[10] A. Iscen, G. Tolias, Y. Avrithis, T. Furon, and O. Chum, "Efficient diffusion on region manifolds: Recovering small objects with compact CNN representations," in *IEEE Conference on Computer Vision and Pattern Recognition*, 2017. [Online]. Available: https://arxiv.org/abs/1611.05113

[11] B. Cao, A. Araujo, and J. Sim, "Unifying deep local and global features for image search," in *European Conference on Computer Vision*, 2020. [Online]. Available: https://arxiv.org/abs/2001.05027

[12] S. Shao, K. Chen, A. Karpur, Q. Cui, A. Araujo, and B. Cao, "Global features are all you need for image retrieval and reranking," in *IEEE/CVF International Conference on Computer Vision*, 2023. [Online]. Available: https://arxiv.org/abs/2308.06954

[13] J. Philbin, O. Chum, M. Isard, J. Sivic, and A. Zisserman, "Object retrieval with large vocabularies and fast spatial matching," in *IEEE Conference on Computer Vision and Pattern Recognition*, 2007, pp. 1–8. [Online]. Available: https://www.robots.ox.ac.uk/~vgg/publications/2007/Philbin07/philbin07.pdf

[14] ——, "Lost in quantization: Improving particular object retrieval in large scale image databases," in *IEEE Conference on Computer Vision and Pattern Recognition*, 2008, pp. 1–8. [Online]. Available: https://www.robots.ox.ac.uk/~vgg/publications/2008/Philbin08/philbin08.pdf

[15] F. Radenović, A. Iscen, G. Tolias, Y. Avrithis, and O. Chum, "Revisiting Oxford and Paris: Large-scale image retrieval benchmarking," in *IEEE Conference on Computer Vision and Pattern Recognition*, 2018. [Online]. Available: https://arxiv.org/abs/1803.11285

[16] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *International Conference on Learning Representations*, 2015. [Online]. Available: https://arxiv.org/abs/1412.6980